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Electronic device

Abstract

An electronic device is disclosed. The electronic device includes a first circuit structure, a first die, a second die, and a third die. The first die is disposed below the first circuit structure. The second die is disposed below the first circuit structure. The third die is disposed above the first circuit structure and electrically connects the first die to the second die. The first die communicates with the second die through the third die.

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Background/Summary

BACKGROUND

1. Field of the Disclosure

(1) The present disclosure relates to an electronic device, and in particular to an electronic package having a plurality of dies.

2. Description of the Related Art

(2) As electronic packages are miniaturized, the number of transistors in a single chip (or processor) increases significantly, and thus the manufacturing yield may decrease due with increased complexity of process. Accordingly, a full-function chip (or processor) may be manufactured by breaking down one processor into multiple partitions (or cores) which are manufactured separately followed by integration and interconnection thereof.

SUMMARY

(3) In some embodiments, an electronic device includes a first circuit structure, a first die, a second die, and a third die. The first die is disposed below the first circuit structure. The second die is disposed below the first circuit structure. The third die is disposed above the first circuit structure and electrically connects the first die to the second die. The first die communicates with the second die through the third die.

(4) In some embodiments, an electronic device includes a computing unit, input/output (I/O) unit, and a bridge unit. The I/O unit is disposed adjacent to the computing unit. The bridge unit is configured to transmit a non-power signal from the computing unit to the I/O unit. The computing unit is configured to receive a power through a backside surface thereof.

(5) In some embodiments, an electronic device includes a memory unit, an input/output (I/O) unit, and a carrier. The computing unit has an active surface facing the memory unit. The I/O unit has an

active surface facing the memory unit. The I/O unit communicates with the computing unit through the memory unit. The carrier is configured to provide a power to the computing unit and dissipate heat from the I/O unit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Aspects of some embodiments of the present disclosure are readily understood from the following detailed description when read with the accompanying figures. It is noted that various structures may not be drawn to scale, and dimensions of the various structures may be arbitrarily increased or reduced for clarity of discussion.
- (2) FIG. 1 is a cross-sectional view of an electronic device in accordance with some embodiments of the present disclosure.
- (3) FIG. 2 is an underside view of an electronic device in accordance with some embodiments of the present disclosure.
- (4) FIG. 3 is a cross-sectional view of an electronic device in accordance with some embodiments of the present disclosure.
- (5) FIG. 4 is a cross-sectional view of an electronic device in accordance with some embodiments of the present disclosure.
- (6) FIG. 5 is a cross-sectional view of an electronic device in accordance with some embodiments of the present disclosure.
- (7) FIG. 6 is a cross-sectional view of an electronic device in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

- (8) Common reference numerals are used throughout the drawings and the detailed description to indicate the same or similar components. Embodiments of the present disclosure will be readily understood from the following detailed description taken in conjunction with the accompanying drawings.
- (9) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to explain certain aspects of the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed or disposed in direct contact, and may also include embodiments in which additional features may be formed or disposed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.
- (10) FIG. 1 is a cross-sectional view of an electronic device **100** in accordance with some embodiments of the present disclosure. The electronic device **100** may include a circuit structure **10**, an encapsulating layer **11**, a circuit structure **12**, dies **13**, **14**, and **15**, and connection elements **16** and **17**.
- (11) The circuit structure (or a carrier, a substrate) **10** is disposed below the die **13** and the die **14**. The circuit structure **10** is disposed below the circuit structure **12**. The circuit structure **10** has a first surface **101** and a second surface **102** opposite thereto. In some embodiments, the circuit structure **10** may include an interposer. In some embodiments, the circuit structure **10** may include, for example, a printed circuit board (PCB), such as a paper-based copper foil laminate, a composite copper foil laminate, or a polymer-impregnated glass-fiber-based copper foil laminate. In some

embodiments, the circuit structure **10** may include a semiconductor substrate including silicon, germanium, or other suitable materials. The circuit structure **10** may be referred as a carrier, or substrate.

(12) The encapsulating layer **11** is disposed on the first surface **101** of the circuit layer **10**. The encapsulating layer **11** is disposed between the circuit layer **10** and the circuit layer **12**. The encapsulating layer **11** has a first surface **111** and a second surface **112** opposite thereto. The first surface **111** faces away from the first surface **101** of the circuit layer **10** and the second surface **112** faces the first surface **101** of the circuit layer **10**. The second surface **112** of the encapsulating layer **11** may be in contact with the first surface **101** of the circuit layer **10**.

(13) In some embodiments, the encapsulating layer **11** may include an epoxy resin including fillers, a molding compound (e.g., an epoxy molding compound or other molding compound), polyimide, a phenolic compound or material, a material including silicone dispersed therein, or a combination thereof.

(14) The circuit structure **12** is disposed over the first surface **111** of the encapsulating layer **11**. The circuit structure **12** is disposed above the circuit structure **10**. The circuit structure **12** is disposed between the dies **13** and **14** and the die **15**. The circuit structure **12** has a first surface **121** and a second surface **122** opposite thereto. The first surface **121** of the circuit structure **12** may face away from the circuit structure **10** and the second surface **122** of the circuit structure **12** may face the circuit structure **10**. In some embodiments, the circuit structure **12** may include, for example, a printed circuit board (PCB), such as a paper-based copper foil laminate, a composite copper foil laminate, or a polymer-impregnated glass-fiber-based copper foil laminate. In some embodiments, the circuit structure **12** may include a semiconductor substrate including silicon, germanium, or other suitable materials.

(15) The die (or a computing unit) **13** may be disposed below the circuit structure **12**. The die **13** may be disposed over the second surface **122** of the circuit structure **12**. The die **13** may be disposed above the circuit structure **10**. The die **13** may be disposed between the circuit structures **10** and **12**. The die **13** may be surrounded by the encapsulating layer **11**. In some embodiments, the die **13** may be encapsulated by the encapsulating layer **11**. The die **13** has a first surface **131** facing the circuit structure **12** and a second surface **132** opposite to the first surface **131** of the die **13**. The first surface **131** may be an active surface, through which the die **13** receives data signals from other components of the electronic device **100** or an external component. The active surface **131** of the die **13** may face the die **15**. The die **13** may include an active region **136**. The active region **136** may include one or more active circuits adjacent to the first surface **131**. The second surface **132** may be a backside surface, through which the die **13** may receive power from other components of the electronic device **100** or an external power source. The die **13** may include one or more conductive pads (not shown) in proximity to, adjacent to, or embedded in and exposed by the first surface **131** or the second surface **132** of the die **13**.

(16) The electronic device **100** may include a plurality of electrical contacts **191** disposed between the first surface **101** of the circuit structure **10** and the second surface **132** of the die **13**. The electrical contacts **191** may attach the die **13** to the circuit structure **10**. The circuit structure may be electrically connected to the die **13** through the electrical contacts **191**. In some embodiments, the electrical contact **191** may include short pillars, solder balls, controlled collapse chip connection (C4) bumps, a ball grid array (BGA), or a land grid array (LGA)

(17) The die **13** may include a plurality of conductive vias (or through-silicon vias) **135** extending through the die **13**. The conductive vias **136** may extend from the second surface **132** to the active region **136** of the die **13**. The conductive vias **135** may attach the die **13** to the circuit structure **10**. The conductive vias **135** may be electrically connected to the conductive vias **107** through the electrical contacts **191**. The power **11** may be transmitted from the circuit structure **10** to the active region **136** of the die **13** through the conductive vias **107**, the electrical contacts **191**, and the conductive vias **135**. In some embodiments, the conductive vias **135** may be formed of metal or

metal alloy. The conductive vias **135** may include metal, such as, copper, gold, silver, aluminum, titanium, tantalum, or the like.

(18) The electronic device **100** may include a plurality of electrical contacts **192** disposed between the first surface **131** of the die **13** and the second surface **122** of the circuit structure **12**. The electrical contacts **192** may attach the die **13** to the circuit structure **12**. The circuit structure **12** may be electrically connected to the die **13** through the electrical contacts **192**. In some embodiments, the electrical contacts **192** may include electroplated short pillars, solder balls, controlled collapse chip connection (C4) bumps, a ball grid array (BGA), or a land grid array (LGA).

(19) In some embodiments, the die **13** may include, for example, a central processing unit (CPU), a microprocessor unit (MPU), a graphics processing unit (GPU), a microcontroller unit (MCU), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), or another type of integrated circuit. In some embodiments, the die **13** may include one or more processing elements and one or more memory elements electrically connected to the processing elements. The processing element(s) and the memory element(s) may be divided from or originate in a monolithic processing unit (e.g., a CPU, a MPU, a GPU, a MCU, an ASIC, or the like). In some embodiments, the processing element may be a CPU chiplet, a MCU chiplet, a GPU chiplet, an ASIC chiplet, or the like. The memory element may be a cache memory. In some embodiments, a wafer node of the processing element is less or smaller than a wafer node of the memory element. In some embodiments, a wafer node of the processing element may lead a wafer node of the memory element by one or more generations. For example, the processing element may be a 5 nm or less node wafer, such as a 3 nm or less node wafer, a 2 nm or less node wafer, or less; and the memory element may be a 20 nm or more node wafer, such as a 28 nm or more node wafer, a 32 nm or more node wafer, or greater.

(20) The die **14** (or input/output (I/O) unit) may be disposed below the circuit structure **12**. The die **14** may be disposed over the second surface **122** of the circuit structure **12**. The die **14** may be disposed above the circuit structure **10**. The die **14** may be disposed between the circuit structures **10** and **12**. The die **14** may be disposed adjacent to the die **13**. The die **14** may be spaced apart from the die **13**. The die **14** may be surrounded by the encapsulating layer **11**. In some embodiments, the die **14** may be encapsulated by the encapsulating layer **11**. The die **14** has a first surface **141** facing the circuit structure **12** and a second surface **142** opposite to the first surface **141** of the die **14**. The first surface **141** may be an active surface, through which the die **14** receives data signals from other components of the electronic device **100** or an external component. The active surface **131** of the die **13** may face the die **15**. The die **14** may include one or more active circuits adjacent to the first surface **141**. The second surface **142** may be a backside surface. The die **14** may include one or more conductive pads (not shown) in proximity to, adjacent to, or embedded in and exposed by the first surface **141** of the die **14**.

(21) The electronic device **100** may include a plurality of electrical contacts **193** disposed between the first surface **141** of the die **14** and the second surface **122** of the circuit structure **12**. The electrical contacts **193** may attach the die **14** to the circuit structure **12**. The circuit structure **12** may be electrically connected to the die **14** through the electrical contacts **193**. In some embodiments, the electrical contacts **193** may include electroplated short pillars, solder balls, controlled collapse chip connection (C4) bumps, a ball grid array (BGA), or a land grid array (LGA).

(22) The die **14** has a lateral surface **143** facing away from the die **13** and a lateral surface **144** facing the die **13**. The lateral surfaces **143** and **144** of die **14** extend from the first surface **141** to the second surface **142** of the die **14**. The die **13** has a lateral surface **133** facing away from the die **14** and a lateral surface **134** facing the die **14**. The lateral surfaces **133** and **134** of die **13** extend from the first surface **131** to the second surface **132** of the die **13**.

(23) The electronic device **100** may further include an adhesive layer **18**. The adhesive layer **18** is disposed between the first surface **101** of the circuit layer **10** and the second surface **142** of the die **14**. The adhesive layer **18** is configured to attach the die **14** to the circuit structure **10**. The adhesive

layer **18** may be electrically isolated and can be referred to as an electrically isolated adhesive layer. With the existence of the adhesive layer **18**, no electrical signal can be transmitted between the die **14** and the circuit structure **10** from the second surface **142** of the die **14**. The adhesive layer **18** may be thermally conductive. The heat can be dissipated from the die **14** to the circuit structure **10** through the adhesive layer **18**.

(24) In some embodiments, the die **14** may include one or more input/output (I/O) units. The I/O units of the die **14** are configured to process the data signals from the die **13**, die **15**, or other components of the electronic device **100**. The I/O units of the die **14** are configured to interconnect the electronic device **100** and an external device (not shown). In some embodiments, the die **14** may be a Serializer/Deserializer (SerDes). The SerDes **14** may be configured to convert data between serial data and parallel interfaces in each direction. The SerDes **14** may provide data transmission over a single line in order to minimize the number of I/O pins and interconnects. In some embodiments, a wafer node of the die **14** may be greater than that of the die **13**.

(25) The die **15** may be disposed above the circuit structure **12**. The die **15** may be disposed over the first surface **121** of the circuit structure **12**. The die **15** may be disposed above the circuit structure **10**. The die **15** has a first surface **151** facing the circuit structure **12** and a second surface **152** opposite to the first surface **151** of the die **15**. The first surface **151** may be an active surface, through which the die **15** receives data signals from other components of the electronic device **100** or an external component. The second surface **152** may be a backside surface. The die **15** may include one or more conductive pads (not shown) in proximity to, adjacent to, or embedded in and exposed by the first surface **151** of the die **15**.

(26) The electronic device **100** may include a plurality of electrical contacts **194** disposed between the first surface **151** of the die **15** and the first surface **121** of the circuit structure **12**. The electrical contacts **194** may attach the die **15** to the circuit structure **12**. The circuit structure **12** may be electrically connected to the die **15** through the electrical contacts **194**. In some embodiments, the electrical contact **194** may include solder balls, controlled collapse chip connection (C4) bumps, a ball grid array (BGA), or a land grid array (LGA).

(27) In some embodiments, the die **15** may be a memory unit (or a data storage unit). The die **15** may include a memory. The die **15** may include dynamic random access memory (DRAM), static random access memory (SRAM), magnetoresistive random access memory (MRAM), flash memory, high bandwidth memory (HBM), or other suitable memory.

(28) The connection element **16** is disposed over the circuit structure **10**. The connection element **16** may be disposed between the circuit structure **10** and the circuit structure **12**. The connection element **16** may be disposed beside the die **13**. The connection element **16** may extend along the lateral surface **133** of the die **13**. The connection element **16** may be surrounded by the encapsulating layer **11**. The connection element **16** may extend between the circuit structures **10** and **12**. The connection element **16** may electrically connect the circuit structure **10** and the circuit structure **12**. The connection element **16** may include a conductive pillar or a conductive via. The connection element **16** may be formed of metal or metal alloy. The connection element **16** may include metal, such as, copper, gold, silver, aluminum, titanium, tantalum, or the like. As shown in FIG. **1**, a plurality of connection elements similar to the connection element **16** may extend along the lateral surface **133** of the die **13** and electrically connect the circuit structure **10** and the circuit structure **12**.

(29) The connection element **17** is disposed over the circuit structure **10**. The connection element **17** may be disposed between the circuit structure **10** and the circuit structure **12**. The connection element **17** may be disposed beside the die **14**. The connection element **17** may extend along the lateral surface **143** of the die **14**. The connection element **17** may be surrounded by the encapsulating layer **11**. The connection element **17** may extend between the circuit structures **10** and **12** through the encapsulating layer **11**. The connection element **17** may electrically connect the circuit structure **10** and the circuit structure **12**. The connection element **17** may include a

conductive pillar or a conductive via. The connection element **17** may be formed of metal or metal alloy. The connection element **17** may include metal, such as copper, gold, silver, aluminum, titanium, tantalum, or the like. As shown in FIG. **1**, a plurality of connection elements similar to the connection element **17** may extend along the lateral surface **143** of the die **14** and electrically connect the circuit structure **10** and the circuit structure **12**.

(30) As shown in FIG. **1**, the circuit structure **10** has a first region **10A** including a plurality of conductive vias (or a non-power vias) **105**. The connection elements **16** are disposed over the first region **10A** and electrically connect the conductive vias **105**. The conductive vias **105** may be aligned with the connection elements **16**. The conductive vias **105** of the circuit structure **10** may be configured to receive a first signal (e.g., data signal, control signal, or non-power signal) **ES1** from the second surface **102** of the circuit structure **10**. The first signal **ES1** is transmitted through a data transmission path (or a non-power signal transmission path or a signal path) **S11**, which is established in the conductive vias **105** and the connection elements **16**. The first signal **ES1** is transmitted along the lateral surface **133** of the die **13**. The first signal **ES1** may be transmitted to the circuit structure **12** through the data transmission path **S11**.

(31) The circuit structure **12** may include a dielectric layer **124**, conductive layers **125**, **126**, **127**, and **128**. The conductive layers **125**, **126**, **127**, or **128** may be surrounded by the dielectric layer **124**. The conductive layer **125**, **126**, **127**, or **128** extends parallel to the first surface **121** of the circuit structure **12**. The conductive layer **125**, **126**, **127**, or **128** may have a portion extending parallel to the first surface **121** of the circuit structure **12**. The conductive layer **125**, **126**, **127**, or **128** may extend parallel to the first surface **151** of the die **15**, the first surface **141** of the die **14**, or the first surface **131** of the die. The conductive layers **125**, **126**, **127**, and **128** may be formed of metal or metal alloy. The conductive layers **125**, **126**, **127**, and **128** may include metal, such as copper, gold, silver, aluminum, titanium, tantalum, or the like.

(32) The conductive layer **125** overlaps the connection element **16** and the die **13** in a direction perpendicular to the first surface **121** of the circuit structure **12**. The conductive layer **125** is configured to provide an interconnection between the connection element **16** and the die **13**. The conductive layer **125** may receive the first signal **ES1** from the connection elements **16**.

Subsequently, the first signal **ES1** is transmitted from the connection element **16** (or a portion of the data transmission path **S11**) to the die **13** (e.g., the active circuits adjacent to the active surface **131** of the die **13**) through a data transmission path **S13**, which is established in the circuit structure **12** (e.g., the conductive layer **125**). The data transmission path **S13** may be further included in the electrical contacts **192** and conductive pads (not shown) on or adjacent to the first surface **131** of the die **13**. The first die **13** is configured to receive the first signal **ES1** through the data transmission paths **S11** and **S13** (can be collectively referred to a non-power transmission path). The non-power signal transmission path **S11** and **S13** is provided along the lateral surface **133** of the die **13** and in the circuit structure **12**. In other words, the circuit structure **12** extends over the lateral surface **133** of the die **13** to transmit the first signal **ES1** to the die **13**.

(33) The die **13** is configured to generate a second signal (e.g., data signal, control signal, or non-power signal) **ES2** in response to the first signal **ES1**. For example, the die **13** may be configured to generate the second signal **ES2** with a data format different from that of the first signal **ES1**, but include substantially the same information. In yet another example, the die may be configured to generate the second signal **ES2** based on the calculation or analysis of the first signal **ES1**. In some embodiments, the die **13** may be configured to transmit the first signal **ES1**, and thus the first signal **ES1** and the second signal **ES2** may be the same.

(34) The conductive layer **126** overlaps the die **13** in a direction perpendicular to the first surface **121** of the circuit structure **12**. The conductive layer **126** overlaps the die **15** in a direction perpendicular to the first surface **121** of the circuit structure **12**. The conductive layer **126** may have a portion extending perpendicular to the first surface **121** of the circuit structure **12**. The conductive layer **126** is configured to provide an interconnection between the die **13** and the die **15**.

(35) The die **15** may communicate with the die **13** through the circuit structure **12**. The second signal **ES2** is transmitted from the die **13** to the die **15** through a data transmission path **S15**, which is established in the circuit structure **12** (e.g., the conductive layer **126**). The second signal **ES2** is transmitted through a data transmission path **S15**, which is established in the circuit structure **12** (e.g., the conductive layer **126**). The data transmission path **S15** may be included in one or more of the electrical contacts **194** and conductive pads (not shown) on or adjacent to the first surface **121** of the circuit structure **12**, the first surface **131** of the die **13**, or the first surface **151** of the die **15**.

(36) The die **15** may include a conductive layer **158**, and. The conductive layer **158** extends parallel to the first surface **121** of the circuit structure **12**. The conductive layer **158** extends parallel to the first surface **151** of the die **15**. The die **15** may electrically connect the die **13** to the die **14**. The die **13** may communicate with the die **14** through the die **15**. The die **15** may be configured to transmit the second signal **ES2** from the die **13** to the die **14**. The second signal **ES2** may be transmitted through a data transmission path **S16**, which is established in the die **15** (e.g., the conductive layer **158**). The second signal **ES2** can be transmitted through the die **15**. The die **15** may be configured to transmit a non-power signal (e.g., the second signal **ES2**) from the die (or the computing unit) **13** to the die (or the I/O unit) **14**. In other words, the die **15** can be referred to as a bridge unit (or a bridge die) which provides an interconnection between a plurality of dies. The die **15** can have a function that cooperates with the dies **13** and **14**. For example, the die **13** can read the data that is stored in the memory of the die **15** or write the memory in the die **15**. The die **14** can read the data that is stored in the die **15**. The die **13** may be configured to access the data in the die **15**. The data stored in the die **15** is configured to be accessed by the die **13**.

(37) The conductive layer **127** overlaps the die **13** in a direction perpendicular to the first surface **121** of the circuit structure **12**. The conductive layer **127** overlaps the die **15** in a direction perpendicular to the first surface **121** of the circuit structure **12**. The conductive layer **127** may have a portion extending perpendicular to the first surface **121** of the circuit structure **12**. The conductive layer **127** is configured to provide an interconnection between the die **13** and the die **15**.

(38) The die **15** may communicate with the die **14** through the circuit structure **12** (e.g., the conductive layer **127**). The die **15** may transmit the second signal **ES2** to the circuit structure **12** (e.g., the conductive layer **127**). The second signal **ES2** may be transmitted to the die **14** through a data transmission path **S17**, which is established in the circuit structure **12** (e.g., the conductive layer **127**). The data transmission path **S17** may be included in one or more of the electrical contacts **194**, one or more of the electrical contacts, and conductive pads (not shown) on or adjacent to the first surface **121** of the circuit structure **12**, the first surface **131** of the die **13**, or the first surface **151** of the die **15**.

(39) The die **14** may receive the second signal **ES2** by the first surface **141** thereof. The die **14** may be configured to transfer the second signal **ES2** to an external device.

(40) The conductive layer **128** overlaps the connection element **17** and the die **14** in a direction perpendicular to the first surface **121** of the circuit structure **12**. The conductive layer **128** is configured to provide an interconnection between the connection element **17** and the die **14**. The conductive layer **128** may receive the second signal **ES2** from the die **14**. Subsequently, the second signal **ES2** is transmitted from the die **14** to the connection element **17** through a data transmission path **S19**, which is established in the circuit structure **12** (e.g., the conductive layer **128**). The data transmission path **S19** may be further included in the electrical contacts **193** and conductive pads (not shown) on or adjacent to the first surface **141** of the die **14**. In other words, the circuit structure **12** extends over the lateral surface **143** of the die **14** to transmit the second signal **ES2** from the die **14**.

(41) The circuit structure **10** may be configured to receive the second signal **ES2** from the circuit structure **12** (e.g., the conductive layer **128**). The circuit structure **10** may include a plurality of conductive vias **106** electrically connects the connection elements **17**. The conductive vias **106** may be aligned with the connection elements **17**. The second signal **ES2** is transmitted through a data

transmission path **S21**, which is established in the connection elements **17** and the conductive vias **106**. The second signal **ES2** is transmitted along the lateral surface **143** of the die **14**. The second signal **ES2** may be transmitted to the circuit structure **10** through the data transmission path **S21**. The circuit structure **10** may be configured to transmit the second signal **ES2** from the second surface **102** of the circuit structure **10**.

(42) Accordingly, the first signal **ES1** is transmitted to the die **13** (e.g., the active surface **131** of the die **13**) through the data transmission paths **S11** and **S13**. The second signal **ES2**, which is associated with the first signal **ES1**, is transmitted from the die **13** through the data transmission paths **S15**, **S16**, **S17**, **S19**, and **S21**. In some embodiments, the second signal **ES2** is transmitted from the die **13** to the die **14** through the data transmission paths **S15**, **S16**, and **S17**. The data transmission paths **S15**, **S16**, and **S17** may be collectively referred as a data transmission path consisting of the signal path from the die **13** to the die **15**, the signal path **S17** from the die **15** to the die **14** and the signal path **S16** in the die **15**. The signal path **S16** connects the signal path **S15** with the signal path **S17**. In some embodiments, the second signal **ES2** is transmitted from the die **14** through the data transmission paths **S19** and **S21**. In some embodiments, the circuit structure **12** is configured to provide the data transmission paths **S11**, **S15**, and/or **S17**.

(43) In some embodiments, the first signal **ES1** is transmitted from the circuit structure **10** to the die **13** through the connection element **16** and the circuit structure **12**. In some embodiments, the die **13** is configured to generate the second signal **ES2** which is transmitted to the circuit structure **10** through the die **15**, the die **14**, the circuit structure **12**, and the connection element **17**.

(44) In some embodiments, the circuit structure **10** may be configured to transmit the first signal **ES1** to the die **13** through the connection element **16** and the circuit structure **12**, and to receive the second signal **ES2** from the die **13** through the die **15**, the die **14**, the circuit structure **12**, and the connection element **17**.

(45) In some embodiments, the direction of the data transmission can be reversed. That is, the circuit structure **10** may be configured to receive an input signal from the second surface **102** through the conductive vias **106**. The input signal can be transmitted to the die **13** through the connection element **17**, the circuit structure **12**, die **14**, and the die **15**. The die **13** may be configured to process the input signal and generate an output signal, which will be transmitted to other components of the electronic device **100** or an external components through the circuit structure **12**, the connection element **16**, and the conductive vias **105** of the circuit structure **10**.

(46) The electronic device **100** may be configured to transmit one or more first signals and one or more second signals associated with the first signals between the dies of the electronic device **100** through the bridge interconnection provided by the circuit structure **12** and the die **15**. The bridge die **15** is disposed at a different elevation than dies **13** and **14** and overlaps the dies **13** and **14** along a Z axis as shown in FIG. 6, such that the distance between the dies **13** and **14** along an X axis as shown in FIG. 6 can be small and the size of the electronic component **100** in a direction along the X-axis can be reduced. Similarly, the size of the electronic component **100** in a direction along a Y-axis (not shown) can be reduced. In an alternative embodiment, the die may not overlap the dies **13** and **14** along the Z axis. The conductive layers **126** and **127** may have a longer portion parallel to the first surface **121** of the circuit structure **12** to overlap the die and dies **13** and **14**, respectively, for providing interconnection between the die **13** and die **15** and between the die **14** and die **15**.

(47) In the present disclosure, the die **15** is configured as a bridge for the communication between the die **13** and the die **14** and configured for a specific function, e.g., data storage. Owing to the integration of a bridge die and a function die, the size of the electronic device **100** in a direction along the X-axis can be reduced. The circuit structure **12** extends over the lateral surface **131** of the die **13** and the lateral surface **141** of the die **14** to transmit the first signal **ES1** to the die **13** and transmit the second signal **ES2** from the die **13**. Furthermore, the circuit structure **12** transmits the first signal **ES1** from the connection element **16** to the die **13**. The circuit structure **12** transmits the second signal **ES2** from the die **14** to the connection element **17**. The circuit structure **12** provides a

horizontal interconnection between the connection elements (e.g., the connections **16** and **17**) and the dies (e.g., the dies **13** and **14**) that are dislocated from the connection elements relative to the first surface **121** of the circuit structure **12**.

(48) The circuit structure **12** is disposed between the dies **13** and **14** and the die **15**. Such an arrangement may be configured to provide interconnection among these dies. Owing to the arrangement of the circuit structure **12**, the vertical data transmission path between the dies **13** and **14** and the die can be reduced, such that the noise and the transmission loss can be reduced.

(49) The circuit structure **10** further has a second region **10B** including a plurality of conductive vias (or power vias) **107**. The die **13** may be disposed over the second region **10B** of the circuit structure and electrically connects with the conductive vias **107**. The second region may be disposed under the die **13**. The second region **10B** may be covered by the die **13**. The conductive vias (or the power vias) **107** may be disposed under the die (or the computing unit) **13**. On the other hands, the first region **10A** may be uncovered by the die **13**. The conductive vias (or the non-power vias) **105** may be uncovered by the die (or the computing unit) **13**. The conductive vias **107** may be electrically connected to the conductive vias **135**. The circuit structure **10** is configured to provide power **P11** to the backside surface **132** of the die **13** through the conductive vias **107**. The die **13** may be configured to receive the power **P11** through the backside surface **132** thereof. A width **W1** of one of the conductive vias **135** may be less than a width **W2** of one of the conductive vias **107**. The power can be provided to the backside surface **132** of the die **13**, rather than the active surface **131** of the die. The power transmission path can be relatively short.

(50) In some embodiments, the electronic device **100** may further include a plurality of electrical contacts or solder balls (not shown) disposed on the second surface **102** of the circuit structure **10** connected to the conductive vias **105**, **106**, and/or **107**. The conductive vias **105**, **106**, and/or **107** may be connected to a printed circuit board, an external device, carrier, or system through the plurality of electrical contacts (or solder balls) on the second surface **102** of the circuit structure **10**. The conductive vias **105**, **106**, and/or **107** may be formed of metal or metal alloy. The conductive vias **105**, **106**, and/or **107** may include metal, such as, copper, gold, silver, aluminum, titanium, tantalum, or the like.

(51) FIG. 2 is an underside view of an electronic device (e.g., the electronic device **100**) in accordance with some embodiments of the present disclosure. As shown in FIG. 2, the circuit structure **10** has the first region **10A** including the plurality of conductive vias **105**, and the second region **10B** including the plurality of conductive vias **107**. The distribution of the conductive vias **105** may define the first region **10A**. The distribution of the conductive vias **107** may define the second region **10B**.

(52) The conductive vias **105** have a pitch **P3**. The conductive vias **107** have a pitch **P2**. The pitch **P2** of the plurality of conductive vias **107** is greater than the pitch **P3** of the plurality of conductive vias **105**. The conductive vias **105** has a width **W3**. The conductive vias **105** has a space **S3**. The pitch **P3** of the conductive vias **105** is equal to the width **W3** plus the space **S3**. The conductive vias **107** has a width **W2**. The conductive vias **107** has a space **S2**. The pitch **P2** of the conductive vias **107** is equal to the width **W2** plus the space **S2**. The width **W2** of the plurality of conductive vias **107** is greater than the width **W3** of the conductive vias **105**. The pitch **P2** of the plurality of conductive vias **107** is greater than the pitch **P3** of the conductive vias **105**.

(53) The circuit structure **10** includes a plurality of sets of conductive vias (e.g., the conductive vias **105** and **107**) with multiple different pitches. The plurality of conductive vias **105** with a finer size are configured to transmit non-power signals (e.g., data signals or control signals). The first region **10A** may be referred to as a data transmission region. The first region may be configured to transmit a non-power signal (e.g., the first signal **ES1**) to the die **13**. The plurality of conductive vias **107** with a greater size are configured to transmit power signals. The second region **10B** may be referred to as a power transmission region. The second region may be configured to transmit a power (e.g., the power **P11**) to the die **13** through the backside surface **132** thereof. The first region

10B is distinct from the first region **10A**. The numbers of the conductive vias **105** may be greater than that of the conductive vias **107**. Alternatively, the numbers of the conductive vias **105** may be less than that of the conductive vias **107**; the numbers of the conductive vias **105** may be the same as that of the conductive vias **107**.

(54) FIG. 3 is a cross-sectional view of an electronic device **200** in accordance with some embodiments of the present disclosure. The electronic device **200** in FIG. 3 is similar to the electronic device **100** in FIG. 1. Therefore, some detailed descriptions may refer to corresponding preceding paragraphs and are not repeated hereinafter for conciseness, with differences therebetween as follows.

(55) As shown in FIG. 3, the circuit structure **10** of the electronic device **200** further has a third region **10C** configured to transfer or dissipate heat from the die **14**. The carrier **10** may be configured to provide a power (e.g., the power **P11**) to the computing unit **13** and dissipate heat from the I/O unit **14**. The third region **10C** may be disposed under the die **14**. The third region includes a thermal via **109**. The thermal via **109** may be disposed under the die **14**. The thermal via **109** is thermally conductive. The thermal via **109** is more thermally conductive than the dielectric material of the circuit structure **10**. The third region **10C** includes more thermal vias similar to the thermal via **109**. A width **W4** of the thermal vias **109** may be greater than the width **W2** of the power via **107**. The thermal vias **109** of the circuit structure **10** can reduce the risk of crack and/or alleviate dishing effect during the polishing process of the circuit structure **10**. The variation of the thickness of the circuit structure **10** from the first surface **101** to the second surface **102** can be reduced.

(56) The adhesive layer **18** may be electrically isolated but thermally conductive. The adhesive layer **18** may include heat dissipation gel or thermal interface material, which may include as silicone, wax, polymer, metal, or other suitable material. The heat of the die **14** may be dissipated to an external environment through the adhesive layer **18** and the thermal via **109**.

(57) In some embodiments, the electronic device **100** may further include a plurality of electrical contacts or solder balls (not shown) disposed on the second surface **102** of the circuit structure **10** connected to the thermal vias **109**. The thermal vias **109** may be connected to a printed circuit board, an external device, carrier, or system through the plurality of electrical contacts (or solder balls) on the second surface **102** of the circuit structure **10**. The conductive vias **109** may be formed of metal or metal alloy. The conductive vias **109** may include metal, such as, copper, gold, silver, aluminum, titanium, tantalum, or the like.

(58) FIG. 4 is a cross-sectional view of an electronic device **300** in accordance with some embodiments of the present disclosure. The electronic device **300** in FIG. 4 is similar to the electronic device **200** in FIG. 3. Therefore, some detailed descriptions may refer to corresponding preceding paragraphs and are not repeated hereinafter for conciseness, with differences therebetween as follows.

(59) The electronic device **300** further includes a circuit structure **52** disposed over the circuit structure **10**. The circuit structure **52** may be disposed below the dies **13** and **14** and the connection elements **16** and **17**. The circuit structure **52** may be configured to electrically connect the dies **13** and **14**, and the connection elements **16** and **17** to the circuit structure **10**. The circuit structure **52** has a first surface **521** and a second surface **522** opposite to the first surface **521** thereof. The first surface **521** of the circuit structure **52** may face away from the circuit structure **10** and the second surface **522** of the circuit structure **52** may face the circuit structure **10**. In some embodiments, the circuit structure **52** may include, for example, a printed circuit board (PCB), such as a paper-based copper foil laminate, a composite copper foil laminate, or a polymer-impregnated glass-fiber-based copper foil laminate. In some embodiments, the circuit structure **52** may include a semiconductor substrate including silicon, germanium, or other suitable materials.

(60) The circuit structure **52** includes conductive layers **524**, **525**, and **526**. The conductive layers **524**, **525**, and **526** may be formed of metal or metal alloy. The conductive layers **524**, **525**, and **526**

may include metal, such as, copper, gold, silver, aluminum, titanium, tantalum, or the like.

(61) The conductive layer **524** may extend from the first surface **521** to the second surface **522** of the circuit structure **52**. The conductive layer **524** may provide interconnection between the conductive vias **105** of the circuit structure **10** and the connection elements **16**. The conductive layer **526** may extend from the first surface **521** to the second surface **522** of the circuit structure **52**. The conductive layer **526** may provide interconnection between the conductive vias **106** of the circuit structure **10** and the connection elements **17**.

(62) The electronic device **300** further includes a passive component (or a capacitor structure) **21** disposed within the circuit structure **10**. In some embodiments, the passive component **21** is accommodated in a cavity defined by the circuit structure **10**. The capacitor **21** structure is attached to a bottom surface of the cavity through an adhesive layer **22**. The capacitor **21** structure is surrounded by an encapsulant **23**, which may fill the cavity. In some embodiments, the encapsulant **23** may include an epoxy resin including fillers, a molding compound (e.g., an epoxy molding compound or other molding compound), polyimide, a phenolic compound or material, a material including silicone dispersed therein, or a combination thereof. The passive component **21** is attached to the circuit structure **52** through a plurality of electrical contacts **195**. In some embodiments, the electrical contact **195** may include solder balls, controlled collapse chip connection (C4) bumps, a ball grid array (BGA), or a land grid array (LGA).

(63) The passive component **21** is disposed below or under the die **13**. The passive component **21** is electrically connected to the die **13** through the circuit structure **52** (e.g., the conductive layer **525**). The passive component **21** may be electrically connected to the conductive vias **107** through the circuit structure **52** (e.g., the conductive layer **525**). The passive component **21** may be configured to receive power from an external power source through the conductive vias **107** of the circuit structure **10**. The passive component **21** is configured to adjust power transmitted to the die **13**. The passive component **21** is configured to stabilize the power transmitted to the die **13**. The passive component **21** is configured to filter noise from the power transmitted to the die **13**. The passive component **21** may include a decoupling capacitor. The passive component **21** may include a capacitor, a surface mount technology (SMT) capacitor device, a ceramic capacitor, a deep trench capacitor (DTC), or the like.

(64) FIG. 5 is a cross-sectional view of an electronic device in accordance with some embodiments of the present disclosure. The electronic device **400** in FIG. 5 is similar to the electronic device **200** in FIG. 3. Therefore, some detailed descriptions may refer to corresponding preceding paragraphs and are not repeated hereinafter for conciseness, with differences therebetween as follows.

(65) The electronic device **400** further includes a circuit (capacitor structure) **31** disposed under the second surface **132** of the die **13**. The circuit **31** may be surrounded by the encapsulating layer **11**. In some embodiments, the circuit **31** may be encapsulated by the encapsulating layer **11**. The circuit **31** may include a thin film capacitor. The circuit **31** is electrically connected to the die **13** and the conductive vias **107**. The circuit **31** may be configured to receive power from an external power source through the conductive vias **107** of the circuit structure **10**. The circuit **31** is configured to adjust power transmitted to the die **13**. The circuit **31** is configured to stabilize the power transmitted to the die **13** and filter noise from the power transmitted to the die **13**. The circuit **31** may include a decoupling capacitor.

(66) FIG. 6 is a cross-sectional view of an electronic device **500** in accordance with some embodiments of the present disclosure. The electronic device **500** in FIG. 6 is similar to the electronic device **100** in FIG. 1. Therefore, some detailed descriptions may refer to corresponding preceding paragraphs and are not repeated hereinafter for conciseness, with differences therebetween as follows.

(67) As shown in FIG. 6, the electronic device **500** further including an encapsulating layer **50** disposed on the first surface **121** of the circuit layer **12**. The encapsulating layer **50** may cover the first surface **121** of the circuit layer **12**. The die **15** may be surrounded by the encapsulating layer

50. The die 15 may be encapsulated by the encapsulating layer 50. The electrical contacts 194 may be surrounded by the encapsulating layer 50. The electrical contacts 194 may be encapsulated by the encapsulating layer 50.

(68) In some embodiments, the encapsulating layer 50 may include an epoxy resin including fillers, a molding compound (e.g., an epoxy molding compound or other molding compound), polyimide, a phenolic compound or material, a material including silicone dispersed therein, or a combination thereof.

(69) Spatial descriptions, such as “above,” “below,” “up,” “left,” “right,” “down,” “top,” “bottom,” “vertical,” “horizontal,” “side,” “higher,” “lower,” “upper,” “over,” “under,” and so forth, are indicated with respect to the orientation shown in the figures unless otherwise specified. It should be understood that the spatial descriptions used herein are for purposes of illustration only, and that practical implementations of the structures described herein can be spatially arranged in any orientation or manner, provided that the merits of embodiments of this disclosure are not deviated from by such an arrangement.

(70) As used herein, the terms “approximately,” “substantially,” “substantial” and “about” are used to describe and account for small variations. When used in conjunction with an event or circumstance, the terms can refer to instances in which the event or circumstance occurs precisely as well as instances in which the event or circumstance occurs to a close approximation. For example, when used in conjunction with a numerical value, the terms can refer to a range of variation less than or equal to $\pm 10\%$ of that numerical value, such as less than or equal to $\pm 5\%$, less than or equal to $\pm 4\%$, less than or equal to $\pm 3\%$, less than or equal to $\pm 2\%$, less than or equal to $\pm 1\%$, less than or equal to $\pm 0.5\%$, less than or equal to $\pm 0.1\%$, or less than or equal to $\pm 0.05\%$. For example, two numerical values can be deemed to be “substantially” the same or equal if a difference between the values is less than or equal to $\pm 10\%$ of an average of the values, such as less than or equal to $\pm 5\%$, less than or equal to $\pm 4\%$, less than or equal to $\pm 3\%$, less than or equal to $\pm 2\%$, less than or equal to $\pm 1\%$, less than or equal to $\pm 0.5\%$, less than or equal to $\pm 0.1\%$, or less than or equal to $\pm 0.05\%$.

(71) Two surfaces can be deemed to be coplanar or substantially coplanar if a displacement between the two surfaces is no greater than 5 μm , no greater than 2 μm , no greater than 1 μm , or no greater than 0.5 μm .

(72) As used herein, the singular terms “a,” “an,” and “the” may include plural referents unless the context clearly dictates otherwise.

(73) As used herein, the terms “conductive,” “electrically conductive” and “electrical conductivity” refer to an ability to transport an electric current. Electrically conductive materials typically indicate those materials that exhibit little or no opposition to the flow of an electric current. One measure of electrical conductivity is Siemens per meter (S/m). Typically, an electrically conductive material is one having a conductivity greater than approximately $10^{4.5}$ S/m, such as at least $10^{4.5}$ S/m or at least $10^{4.6}$ S/m. The electrical conductivity of a material can sometimes vary with temperature. Unless otherwise specified, the electrical conductivity of a material is measured at room temperature.

(74) Additionally, amounts, ratios, and other numerical values are sometimes presented herein in a range format. It is to be understood that such range format is used for convenience and brevity and should be understood flexibly to include numerical values explicitly specified as limits of a range, but also to include all individual numerical values or sub-ranges encompassed within that range as if each numerical value and sub-range is explicitly specified.

(75) While the present disclosure has been described and illustrated with reference to specific embodiments thereof, these descriptions and illustrations are not limiting. It should be understood by those skilled in the art that various changes may be made and equivalents may be substituted without departing from the true spirit and scope of the present disclosure as defined by the appended claims. The illustrations may not be necessarily drawn to scale. There may be

distinctions between the artistic renditions in the present disclosure and the actual apparatus due to manufacturing processes and tolerances. There may be other embodiments of the present disclosure which are not specifically illustrated. The specification and drawings are to be regarded as illustrative rather than restrictive. Modifications may be made to adapt a particular situation, material, composition of matter, method, or process to the objective, spirit and scope of the present disclosure. All such modifications are intended to be within the scope of the claims appended hereto. While the methods disclosed herein have been described with reference to particular operations performed in a particular order, it will be understood that these operations may be combined, sub-divided, or re-ordered to form an equivalent method without departing from the teachings of the present disclosure. Accordingly, unless specifically indicated herein, the order and grouping of the operations are not limitations of the present disclosure.

Claims

1. An electronic device, comprising: a first circuit structure; a first die disposed below the first circuit structure; a second die disposed below the first circuit structure; and a third die disposed above the first circuit structure and electrically connecting the first die to the second die, wherein the first die communicates with the second die through the third die, wherein the first die includes an active surface facing the first circuit structure and a backside surface opposite to the active surface of the first die, wherein the first die is configured to receive a power through the backside surface thereof.
2. The electronic device of claim 1, wherein the third die is a data storage unit, wherein the first die is configured to access data in the third die.
3. The electronic device of claim 2, wherein the second die is a Serializer/Deserializer (SerDes).
4. The electronic device of claim 1, wherein a non-power signal transmission path is provided along a lateral surface of the first die and in the first circuit structure, and wherein a non-power signal is transmitted to an active circuit adjacent to the active surface of the first die through the non-power signal transmission path.
5. The electronic device of claim 1, further comprising: a second circuit structure having a first region and a second region distinct from the first region, wherein the first region of the second circuit structure is configured to transmit a non-power signal to the first die through the first circuit structure, and wherein the second region of the second circuit structure is configured to transmit the power to the first die through the backside surface of the first die.
6. The electronic device of claim 5, wherein the first region is uncovered by the first die, and wherein the second region is disposed under the first die.
7. The electronic device of claim 6, wherein the first die includes a first conductive via extending through the first die, and wherein the second region comprises a second conductive via electrically connected to the first conductive via.
8. The electronic device of claim 7, wherein a width of the first conductive via is less than that of the second conductive via.
9. The electronic device of claim 5, wherein the second circuit structure further has a third region configured to dissipate heat from the second die.
10. The electronic device of claim 1, wherein a data transmission path consists of a first signal path from the first die to the third die, a second signal path from the third die to the second die, and a third signal path in the third die, wherein the third signal path connects the first signal path with the second signal path.
11. The electronic device of claim 1, wherein the second die is disposed adjacent to the first die, wherein the third die is configured to transmit a non-power signal from the first die to the second die, wherein data stored in the third die is configured to be accessed by the first die.
12. The electronic device of claim 11, wherein the first circuit structure is configured to provide a

first data transmission path for transmitting a first non-power signal to the first die; wherein the first circuit structure is configured to provide a second data transmission path for transmitting a second non-power signal between the first die and the third die; and wherein the first circuit structure is configured to provide a third data transmission path for transmitting a third non-power signal between the third die and the second die.

13. The electronic device of claim 11, further comprising: a second circuit structure comprising a power via and a non-power via, wherein the power via is disposed under the first die, and the non-power via is uncovered by the first die.

14. The electronic device of claim 13, wherein a width of the power via is greater than that of the non-power via.

15. The electronic device of claim 13, wherein a pitch of the power via is greater than that of the non-power via.

16. The electronic device of claim 13, wherein the second circuit structure further comprises a thermal via disposed under the second die, wherein a width of the thermal via is greater than that of the power via.

17. The electronic device of claim 1, wherein the second die has an active surface facing the third die and communicates with the first die through the third die; and wherein the electronic device comprises a carrier configured to provide the power to the first die and dissipate heat from the second die.

18. The electronic device of claim 17, further comprising: a passive component disposed under the first die and configured to stabilize the power and filter noise from the power.

19. The electronic device of claim 17, further comprising: an encapsulating layer encapsulating the first die and the second die, wherein the first die comprises a circuit disposed under the backside surface of the first die, and the circuit is configured to stabilize the power and filter noise from the power, and wherein the circuit is encapsulated by the encapsulating layer.
